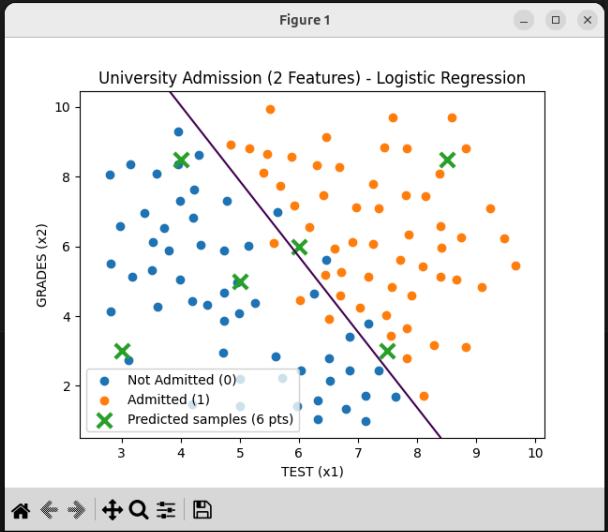
HW6 > WilliamsBenjamin_HW6 > Exercise1 > exercise1.py > ...

```python
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
from sklearn.model_selection import train_test_split
from sklearn.linear_model import LogisticRegression
from sklearn.metrics import accuracy_score, confusion_matrix

df = pd.read_csv("university_admissions_2features.csv")

X = df.iloc[:, 0:2].to_numpy(dtype=float)
y = df.iloc[:, 2].to_numpy()

X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.25, random_state=7, stratify=y
)

model = LogisticRegression(solver="lbfgs", random_state=7)
model.fit(X_train, y_train)

y_pred = model.predict(X_test)
print("Accuracy:", accuracy_score(y_test, y_pred))
print("Confusion Matrix:\n", confusion_matrix(y_test, y_pred))

samples = np.array([
    [4.0, 8.5],
    [7.5, 3.0],
    [6.0, 6.0],
    [3.0, 3.0],
    [8.5, 8.5],
    [5.0, 5.0]
], dtype=float)

sample_preds = model.predict(samples)
print("\nPredicted samples (x1, x2) -> class:")
for s, p in zip(samples, sample_preds):
    print(s.tolist(), "->", int(p))

plt.figure()

m0 = y == 0
m1 = y == 1
plt.scatter(X[m0, 0], X[m0, 1], label="Not Admitted (0)")
plt.scatter(X[m1, 0], X[m1, 1], label="Admitted (1)")

x_min, x_max = X[:, 0].min() - 0.5, X[:, 0].max() + 0.5
y_min, y_max = X[:, 1].min() - 0.5, X[:, 1].max() + 0.5
xx, yy = np.meshgrid(np.linspace(x_min, x_max, 300), np.linspace(y_min, y_max, 300))
grid = np.c_[xx.ravel(), yy.ravel()]
probs = model.predict_proba(grid)[:, 1].reshape(xx.shape)
plt.contour(xx, yy, probs, levels=[0.5])

plt.scatter(samples[:, 0], samples[:, 1], marker="x", s=120, linewidths=3, label="Predicted samples (6 pts)")

plt.title("University Admission (2 Features) - Logistic Regression")
plt.xlabel("TEST (x1)")
plt.ylabel("GRADES (x2)")
plt.legend()
plt.show()
```

PROBLEMS OUTPUT DEBUG CONSOLE PORTS

No problems have been detected in the workspace.



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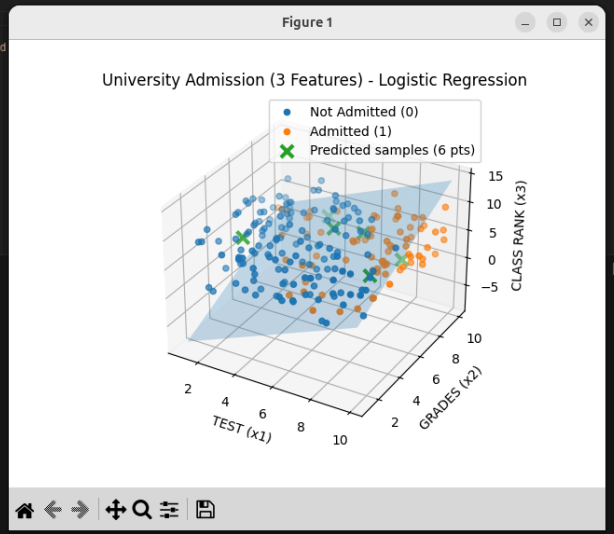
```
/usr/local/bin/python3.14 /home/ben/Documents/Development/MachineLearning/HW/HW6/WilliamsBenjamin_HW6/Exercise1/exercise1.py
ben@ben-System-Product-Name:~/Documents/Development/MachineLearning/HW$ /usr/local/bin/python3.14 /home/ben/Documents/Development/MachineLearning/HW/HW6/WilliamsBenjamin_HW6/Exercise1/exercise1.py
Traceback (most recent call last):
  File "/home/ben/Documents/Development/MachineLearning/HW/HW6/WilliamsBenjamin_HW6/Exercise1/exercise1.py", line 8, in <module>
    df = pd.read_csv("university_admissions_2features.csv")
  File "/home/ben/.local/lib/python3.14/site-packages/pandas/io/parsers/readers.py", line 873, in read_csv
    return _read(filepath_or_buffer, kwds)
  File "/home/ben/.local/lib/python3.14/site-packages/pandas/io/parsers/readers.py", line 300, in _read
    parser = TextFileReader(filepath_or_buffer, **kwds)
  File "/home/ben/.local/lib/python3.14/site-packages/pandas/io/parsers/readers.py", line 1645, in __init__
    self._engine = self._make_engine(f, self.engine)
                   ~~~~~~~~~~~~~~~~~^^^^^^^^^^^^^^^^
  File "/home/ben/.local/lib/python3.14/site-packages/pandas/io/parsers/readers.py", line 1904, in _make_engine
    self.handles = get_handle(
                   ~~~~~~~~~~^
        f,
        ^^
    ...<6 lines>...
        storage_options=self.options.get("storage_options", None),
        ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
    )
    ^
  File "/home/ben/.local/lib/python3.14/site-packages/pandas/io/common.py", line 926, in get_handle
    handle = open(
        handle,
    ...<3 lines>...
        newline="",
    )
FileNotFoundError: [Errno 2] No such file or directory: 'university_admissions_2features.csv'
ben@ben-System-Product-Name:~/Documents/Development/MachineLearning/HW$ cd HW5/WilliamsBenjamin_HW5/Exercise1
ben@ben-System-Product-Name:~/Documents/Development/MachineLearning/HW/HW5/WilliamsBenjamin_HW5/Exercise1$ ls
exercise1.py  image.png  vehicles.csv  vehicles.names
ben@ben-System-Product-Name:~/Documents/Development/MachineLearning/HW/HW5/WilliamsBenjamin_HW5/Exercise1$ cd HW6/WilliamsBenjamin_HW6/Exercise1
bash: cd: HW6/WilliamsBenjamin_HW6/Exercise1: No such file or directory
ben@ben-System-Product-Name:~/Documents/Development/MachineLearning/HW/HW5/WilliamsBenjamin_HW5/Exercise1$ cd  ../../..
ben@ben-System-Product-Name:~/Documents/Development/MachineLearning/HW$ cd HW6
ben@ben-System-Product-Name:~/Documents/Development/MachineLearning/HW/HW6$ cd WilliamsBenjamin_HW6/
ben@ben-System-Product-Name:~/Documents/Development/MachineLearning/HW/HW6/WilliamsBenjamin_HW6$ cd Exercise
bash: cd: Exercise: No such file or directory
ben@ben-System-Product-Name:~/Documents/Development/MachineLearning/HW/HW6/WilliamsBenjamin_HW6$ cd Exercise
bash: cd: Exercise: No such file or directory
ben@ben-System-Product-Name:~/Documents/Development/MachineLearning/HW/HW6/WilliamsBenjamin_HW6$ cd Exercise1
ben@ben-System-Product-Name:~/Documents/Development/MachineLearning/HW/HW6/WilliamsBenjamin_HW6/Exercise1$ /usr/local/bin/python3.14 /home/ben/Documents/Development/MachineLearning/HW/HW6/WilliamsBenjamin_HW6/Exercise1/exercise1.py
Accuracy: 0.9642857142857143
Confusion Matrix:
 [[14  0]
 [ 1 13]]

Predicted samples (x1, x2) -> class:
[4.0, 8.5] -> 0
[7.5, 3.0] -> 1
[6.0, 6.0] -> 1
[3.0, 3.0] -> 0
[8.5, 8.5] -> 1
[5.0, 5.0] -> 0
```

HW6 > WilliamsBenjamin_HW6 > Exercise2 > exercise2.py > ...

```python
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
from sklearn.model_selection import train_test_split
from sklearn.linear_model import LogisticRegression
from sklearn.metrics import accuracy_score, confusion_matrix

df = pd.read_csv("university_admissions_3features.csv")

X = df.iloc[:, 0:3].to_numpy(dtype=float)
y = df.iloc[:, 3].to_numpy()

X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.25, random_state=7, stratify=y
)

model = LogisticRegression(solver="lbfgs", random_state=7)
model.fit(X_train, y_train)

y_pred = model.predict(X_test)
print("Accuracy:", accuracy_score(y_test, y_pred))
print("Confusion Matrix:\n", confusion_matrix(y_test, y_pred))

samples = np.array([
    [8.5, 8.0, 2.0],
    [6.0, 6.0, 8.5],
    [4.0, 9.0, 4.0],
    [9.0, 4.0, 6.5],
    [3.0, 3.0, 9.0],
    [7.0, 7.0, 7.0]
], dtype=float)

sample_preds = model.predict(samples)
print("\nPredicted samples (x1, x2, x3) -> class:")
for s, p in zip(samples, sample_preds):
    print(s.tolist(), "->", int(p))

fig = plt.figure()
ax = fig.add_subplot(111, projection="3d")

m0 = y == 0
m1 = y == 1
ax.scatter(X[m0, 0], X[m0, 1], X[m0, 2], label="Not Admitted (0)")
ax.scatter(X[m1, 0], X[m1, 1], X[m1, 2], label="Admitted (1)")

ax.scatter(samples[:, 0], samples[:, 1], samples[:, 2], marker="x", s=90, linewidths=3, label="Predicted

x1_min, x1_max = X[:, 0].min(), X[:, 0].max()
x2_min, x2_max = X[:, 1].min(), X[:, 1].max()
xx, yy = np.meshgrid(np.linspace(x1_min, x1_max, 30), np.linspace(x2_min, x2_max, 30))

w1, w2, w3 = model.coef_[0]
b = model.intercept_[0]
zz = (-b - w1 * xx - w2 * yy) / (w3 + 1e-12)

ax.plot_surface(xx, yy, zz, alpha=0.25)

ax.set_title("University Admission (3 Features) - Logistic Regression")
ax.set_xlabel("TEST (x1)")
ax.set_ylabel("GRADES (x2)")
```

PROBLEMS OUTPUT DEBUG CONSOLE PORTS

No problems have been detected in the workspace.



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```
ben@ben-System-Product-Name:~/Documents/Development/MachineLearning/HW$ /usr/local/bin/python3.14
 /home/ben/Documents/Development/MachineLearning/HW/HW6/WilliamsBenjamin_HW6/Exercise1/exercise1.
py
  File "/home/ben/.local/lib/python3.14/site-packages/pandas/io/parsers/readers.py", line 300, in
 _read
    parser = TextFileReader(filepath_or_buffer, **kwds)
  File "/home/ben/.local/lib/python3.14/site-packages/pandas/io/parsers/readers.py", line 1645, i
n __init__
    self._engine = self._make_engine(f, self.engine)
                   ~~~~~~~~~~~~~~~~~~^^^^^^^^^^^^^^^^
  File "/home/ben/.local/lib/python3.14/site-packages/pandas/io/parsers/readers.py", line 1904, i
n _make_engine
    self.handles = get_handle(
                   ~~~~~~~~~~^
        f,
        ^^
    ...<6 lines>...
        storage_options=self.options.get("storage_options", None),
        ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
    )
    ^
  File "/home/ben/.local/lib/python3.14/site-packages/pandas/io/common.py", line 926, in get_hand
le
    handle = open(
        handle,
    ...<3 lines>...
        newline="",
    )
FileNotFoundError: [Errno 2] No such file or directory: 'university_admissions_2features.csv'
ben@ben-System-Product-Name:~/Documents/Development/MachineLearning/HW$ cd HW5/WilliamsBenjamin_H
W5/Exercise1
ben@ben-System-Product-Name:~/Documents/Development/MachineLearning/HW/HW5/WilliamsBenjamin_HW5/E
xercise1$ ls
exercise1.py  image.png  vehicles.csv  vehicles.names
ben@ben-System-Product-Name:~/Documents/Development/MachineLearning/HW/HW5/WilliamsBenjamin_HW5/E
xercise1$ cd HW6/WilliamsBenjamin_HW6/Exercise1
bash: cd: HW6/WilliamsBenjamin_HW6/Exercise1: No such file or directory
ben@ben-System-Product-Name:~/Documents/Development/MachineLearning/HW/HW5/WilliamsBenjamin_HW5/E
xercise1$ cd  ../../..
ben@ben-System-Product-Name:~/Documents/Development/MachineLearning/HW$ cd HW6
ben@ben-System-Product-Name:~/Documents/Development/MachineLearning/HW/HW6$ cd WilliamsBenjamin_H
W6/
ben@ben-System-Product-Name:~/Documents/Development/MachineLearning/HW/HW6/WilliamsBenjamin_HW6$
cd Exercise
bash: cd: Exercise: No such file or directory
ben@ben-System-Product-Name:~/Documents/Development/MachineLearning/HW/HW6/WilliamsBenjamin_HW6$
cd Exercise
bash: cd: Exercise: No such file or directory
ben@ben-System-Product-Name:~/Documents/Development/MachineLearning/HW/HW6/WilliamsBenjamin_HW6$
cd Exercise1
ben@ben-System-Product-Name:~/Documents/Development/MachineLearning/HW/HW6/WilliamsBenjamin_HW6/E
xercise1$ /usr/local/bin/python3.14 /home/ben/Documents/Development/MachineLearning/HW/HW6/Willia
msBenjamin_HW6/Exercise1/exercise1.py
Accuracy: 0.9642857142857143
Confusion Matrix:
 [[14  0]
 [ 1 13]]

Predicted samples (x1, x2) -> class:
[4.0, 8.5] -> 0
[7.5, 3.0] -> 1
[6.0, 6.0] -> 1
[3.0, 3.0] -> 0
[8.5, 8.5] -> 1
[5.0, 5.0] -> 0
ben@ben-System-Product-Name:~/Documents/Development/MachineLearning/HW/HW6/WilliamsBenjamin_HW6/E
xercise1$ cd ../Exercise2
ben@ben-System-Product-Name:~/Documents/Development/MachineLearning/HW/HW6/WilliamsBenjamin_HW6/E
xercise2$ ls
exercise2.py  university_admissions_3features.csv
ben@ben-System-Product-Name:~/Documents/Development/MachineLearning/HW/HW6/WilliamsBenjamin_HW6/E
xercise2$ /usr/local/bin/python3.14 /home/ben/Documents/Development/MachineLearning/HW/HW6/Willia
msBenjamin_HW6/Exercise2/exercise2.py
Accuracy: 0.8727272727272727
Confusion Matrix:
 [[35  1]
 [ 6 13]]

Predicted samples (x1, x2, x3) -> class:
[8.5, 8.0, 2.0] -> 1
[6.0, 6.0, 8.5] -> 0
[4.0, 9.0, 4.0] -> 1
[9.0, 4.0, 6.5] -> 0
[3.0, 3.0, 9.0] -> 0
[7.0, 7.0, 7.0] -> 0
```